QUNFENG JIANG

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RESEARCH INTERESTS

Exoplanets; planetary formation and evolution; orbital dynamics; planetary atmospheres.

EDUCATION

The University of Hong Kong, *MPhil in Planetary Sciences* 2024-2026
Supervisor: Prof. Man Hoi Lee
Research: Circumbinary planetary dynamics, free eccentricity, and the orbital evolution of Pluto's small satellites.

Fudan University, *BSc in Physics* 2019-2023
Supervisor: Prof. Cosimo Bambi
Research: X-ray spectroscopy of black hole binaries and tests of general relativity.

RESEARCH EXPERIENCE

Independent Researcher July 2026–Present
Shanghai, China
Collaborators: Prof. Juliette Becker; Dr. Gregory J. Gilbert
Initiated collaborations and led independent projects in exoplanet dynamics and statistical inference.

- **Neptunian desert:** Tested whether cumulative XUV exposure unifies desert boundaries across stellar types using 1,850 Kepler/TESS planets, stellar activity models, and statistical null tests; sole-author paper published in The Astronomical Journal.
- **Hot Jupiter formation:** With Juliette Becker, combined REBOUND/REBOUNDx simulations and tidal calculations to test whether lower-mass giants can undergo high-eccentricity migration while preserving inner planets; manuscript in preparation.
- **Eccentricity dependence:** With Gregory Gilbert, used hierarchical Bayesian models of transit-derived posteriors to investigate eccentricity dependence on orbital period and observed multiplicity across super-Earth and sub-Neptune populations.
- **Eccentricity and prebiotic delivery:** Designed controlled N-body experiments showing that planetary eccentricity can enhance inward comet transport while suppressing low-velocity impacts and prebiotic-molecule survival; sole-author manuscript submitted.
- **π Mensae orbital dynamics:** Combined radial-velocity modeling with three-dimensional N-body integrations to constrain candidate planet d's eccentricity, secular excitation, and viable orbital configurations.
- **Circumbinary planet searches:** Reassessed apsidal-motion planet claims using TESS eclipse timing; found no significant third-body signal after accounting for stellar tidal systematics.

Research Visitor, Earth 2.0 Mission Science Team, Shanghai Astronomical Observatory June–July 2026
Host: Prof. Jian Ge

- Analyzed unusual Kepler transit signals through transit light-curve and timing analysis, including transit timing variations (TTVs), and tested resonant interpretations with N-body simulations and long-term stability analysis.

MPhil Researcher, The University of Hong Kong 2024–2026
Supervisor: Prof. Man Hoi Lee

- Measured free eccentricities of circumbinary planets with N-body integrations and FFTs; proposed a stability-filter hypothesis for their dynamically cold orbits based on K-S analysis.
- Tested an early in-situ formation hypothesis for Pluto's small satellites by combining free-eccentricity constraints with tidal-evolution simulations.

Research Assistant, California Institute of Technology 2023

- Analyzed atmospheric methane and climate time series using satellite observations and reanalysis data, incorporating radiative-transfer methods in the interpretation of atmospheric variability.

Summer Undergraduate Research Fellow, California Institute of Technology 2022

- Modeled NuSTAR X-ray spectra of the black hole binary H 1743-322 in XSPEC, identifying evidence for a mismatch between the inner accretion-disk inclination and the binary orbital plane.

Undergraduate Research Assistant, Fudan University 2020–2021

• Modeled X-ray binary spectra to test relativistic gravity and constrain departures from the Kerr spacetime, contributing to a second-author publication in Physical Review D.

PUBLICATIONS AND MANUSCRIPTS

Refereed Publications and Accepted Papers

1. **Jiang, Q.** (2026). "Does Cumulative XUV Unify the Neptunian Desert?" *The Astronomical Journal*, 172, 306. doi:10.3847/1538-3881/aeb2fb | arXiv:2608.21285
2. **Jiang, Q.** (2026). "An Outer Giant Excites and Constrains the Orbit of π Mensae d." Accepted for publication in *The Astronomical Journal*. arXiv:2609.06682
3. Yu, Z., **Jiang, Q.**, Abdikamalov, A. B., et al. (2021). "Constraining the Konoplya-Rezzolla-Zhidenko deformation parameters. II. Limits from stellar-mass black hole X-ray data." *Physical Review D*, 104, 084035. doi:10.1103/PhysRevD.104.084035

Submitted Manuscripts and Preprints

4. **Jiang, Q.**, & Lee, M. H. (2026). "A Stability Filter Hypothesis for Circumbinary Planets." Revised and resubmitted to *The Astrophysical Journal*.
5. **Jiang, Q.** (2026). "Planetary Eccentricity Enhances Inward Comet Transport but Suppresses Gentle Impacts." Under review at *The Astronomical Journal*. arXiv:2610.01053
6. **Jiang, Q.** (2026). "Stellar tidal systematics in apsidal-motion searches for circumbinary planets: CH Ind and SW CMa." Preprint. arXiv:2608.13269

Manuscripts in Preparation

7. **Jiang, Q.**, & Lee, M. H. "A Dynamical Challenge to Early In Situ Formation in the Pluto–Charon System."
8. **Jiang, Q.** "Lower-Mass Giants Can Spare Inner Planets during High-Eccentricity Migration."

SELECTED PRESENTATIONS

"A Stability-Filter Hypothesis for Circumbinary Planets." 57th AAS Division on Dynamical Astronomy Meeting, 2026. Poster

"A Planet with Two Suns." Earth 2.0 Science Seminar Series and Purple Mountain Observatory, April 2026. Invited seminars

"What Shapes the Orbits of Circumbinary Planets?" Nanjing University Exoplanet Group Seminar, April 2026. Invited seminar

"Dynamical Signatures of Formation and Evolution: Free Eccentricity Analysis of Pluto's Small Satellites." EPSC-DPS Joint Meeting, 2025

"The Mismatch between the Inner-disk and the Orbital Inclination in the Black Hole X-ray Binary H 1743-322." 241st Meeting of the American Astronomical Society, 2023

TECHNICAL SKILLS

Dynamical modeling: N-body integration; REBOUND, REBOUNDx, and SWIFT; resonant dynamics; tidal evolution; stability analysis.

Observational and statistical methods: Kepler and TESS photometry; transit timing variations (TTVs); eclipse timing; radial-velocity modeling; Bayesian and hierarchical inference; MCMC, nested sampling, and K-S tests.

Computing: Python, NumPy, SciPy, pandas, Astropy, Matplotlib; C++; Fortran; XSPEC; Unix and Linux; LaTeX.

TEACHING EXPERIENCE

Teaching Assistant, The University of Hong Kong 2024-2025

The Blue Planet - introductory undergraduate Earth science.

Teaching Assistant, Fudan University 2022-2023

General Physics: Mechanics - introductory undergraduate physics.

AWARDS AND FELLOWSHIPS

HKU Postgraduate Scholarship, 2024-2026.

Chun-Tsung Scholar, CURE Endowment, 2023.

Caltech Summer Undergraduate Research Fellowship, 2022; USD 7,000.